Windows Admin Logon Task Deployment

Technical Reference & Step-by-Step Implementation Guide

To run an external application as an administrator at every non-administrator user logon—while bypassing User Account Control (UAC) prompts and restriction errors like *"This App Has Been Blocked by Your System Administrator"*—the recommended deployment strategy involves a scheduled task running under the `SYSTEM` account with highest privileges.

Step 1: Create the Execution Script

Save the following PowerShell script to a secure, system-accessible directory (e.g., `C:\Scripts\LaunchApp.ps1`). This handles path verification and execution safety.

```
# C:\Scripts\LaunchApp.ps1
$AppPath = "C:\Path\To\YourApplication.exe"
$WorkingDirectory = [System.IO.Path]::GetDirectoryName($AppPath)

if (Test-Path -Path $AppPath) {
    Start-Process -FilePath $AppPath -WorkingDirectory $WorkingDirectory
} else {
    Write-Error "Target application not found at: $AppPath"
}
```

> **Security Note**
> Ensure that standard non-admin users do not have write permissions to the destination folder where the script and target executable reside to prevent privilege escalation vulnerabilities.

Step 2: Deploy the Scheduled Task via PowerShell

Run the deployment script below **as an Administrator** on the target machine. It provisions a system-level task triggered at any user logon.

```
$TaskName       = "ElevatedAppLogonLauncher"
$ScriptPath     = "C:\Scripts\LaunchApp.ps1"
$TaskDescription = "Launches authorized software as administrator at user logon."

$Action = New-ScheduledTaskAction -Execute "powershell.exe" `
    -Argument "-ExecutionPolicy Bypass -WindowStyle Hidden -File `"$ScriptPath`""

$Trigger = New-ScheduledTaskTrigger -AtLogOn

$Principal = New-ScheduledTaskPrincipal -UserId "NT AUTHORITY\SYSTEM" `
    -LogonType ServiceAccount `
    -RunLevel Highest

$Settings = New-ScheduledTaskSettingsSet `
    -AllowStartIfOnBatteries `
    -DontStopIfGoingOnBatteries `
    -StartWhenAvailable `
    -MultipleInstances IgnoreNew

Register-ScheduledTask -TaskName $TaskName `
    -Action $Action -Trigger $Trigger -Principal $Principal `
    -Settings $Settings -Description $TaskDescription -Force
```

> **Why This Bypasses Restrictions**
> Configuring the task principal to run under `NT AUTHORITY\SYSTEM` with `-RunLevel Highest` executes the process outside restricted user token bounds, effectively bypassing software restriction policies, AppLocker, and UAC block dialogs seamlessly.